KEYSIGHT

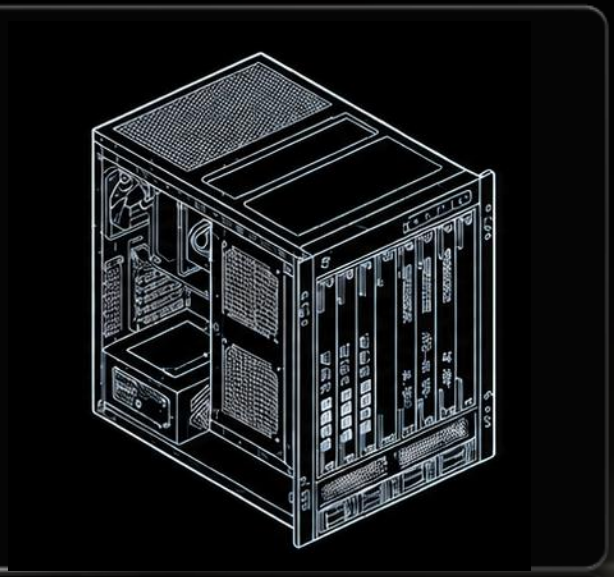
Scaling AI: Design to Deployment Continuum

Validating AI Infrastructure at Scale

Ram Periakaruppan
VP/GM

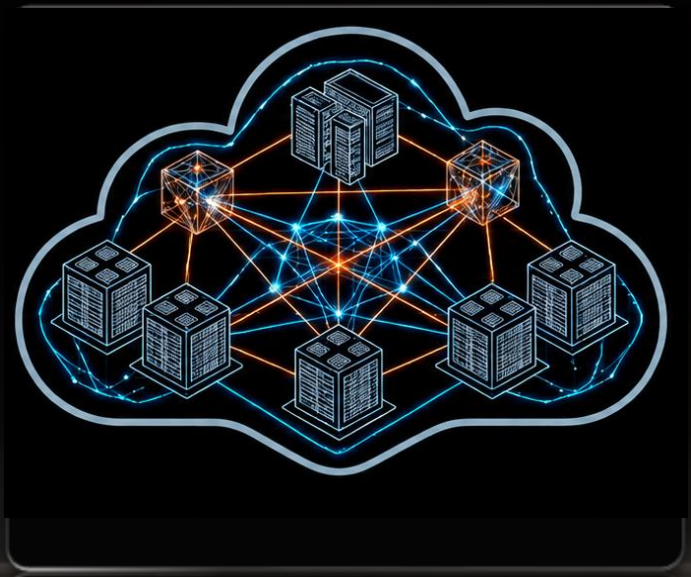
30 Years of Data Center Evolution

Internet Era
(1990s-2000s)



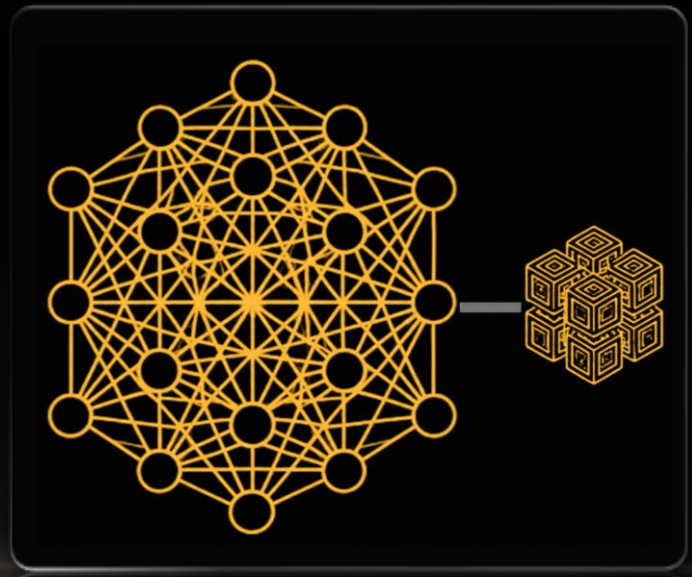
Monolithic Chassis
Single system image
Isolated workloads

Hyperscale Era
(2010s-2020s)



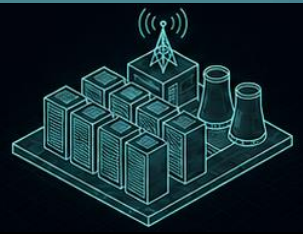
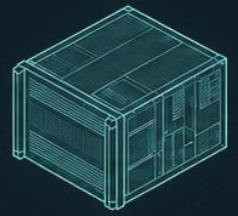
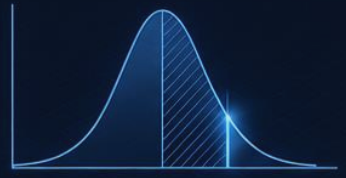
Hyperscale Cloud
Decentralized scaling
Standardized components

AI Era
(2020s-Present)



Back End Network
Synchronized AI Fabric
Hyper optimization

Data Center Architectural Evolution

	Internet Era (1990s-2000s)	Hyperscale Era (2010s-2020s)	AI Era (2020s-Present)
Deployment	 On-Premise	 Virtual, Central Clouds	 Specialized Clusters
Architecture	 Large Integrated Chassis	 Distributed Simple COTS Boxes	 Synchronized Distributed System
Resiliency	 Fault Tolerance Strategy	 N+1 Redundancy	 Weakest Link Failure Mode
Performance	 Performance Limited	 P95 Performance	 Tail Latency Performance

KEYSIGHT

The AI Era Breaks the Hyperscale Rules

We have moved past fault-tolerant commodity scaling

The new paradigm relies on highly synchronized, specialized clusters where **weakest-link failures mean every packet, link, node, and nanosecond counts**



KEYSIGHT

Hyper Optimization

Optimizing For Value vs. Optimizing For Performance



Consumer Car | Traditional Data Centers		Formula 1 | AI Systems	
Architecture	Modular and standardized	Architecture	Bespoke and Unified: a highly specialized
Margins	Standard tolerances: wider margins for standard performance	Margins	Minimal tolerances: narrow margins at peak performance

Where a race is won or lost

Driving comfort

Pit crew

Driver

Fan

Winning a race is a team sport

Driving comfort

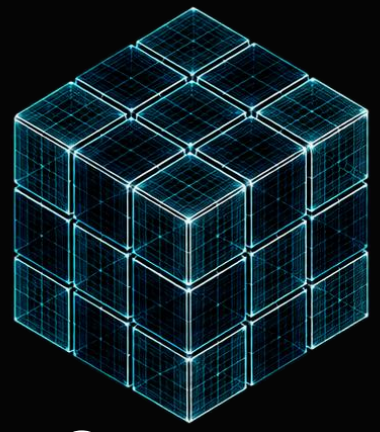
Pit crew

Driver

Fan

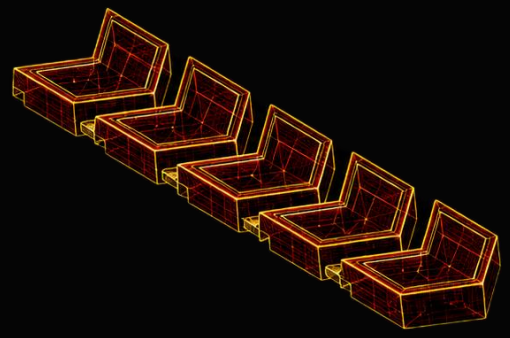
The Continuum – Speeding up Design to Deployment

Tenets of AI Design



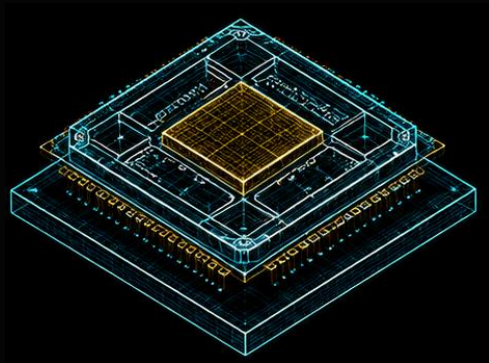
System Thinking

Quality



Design to Deployment Continuum

Velocity

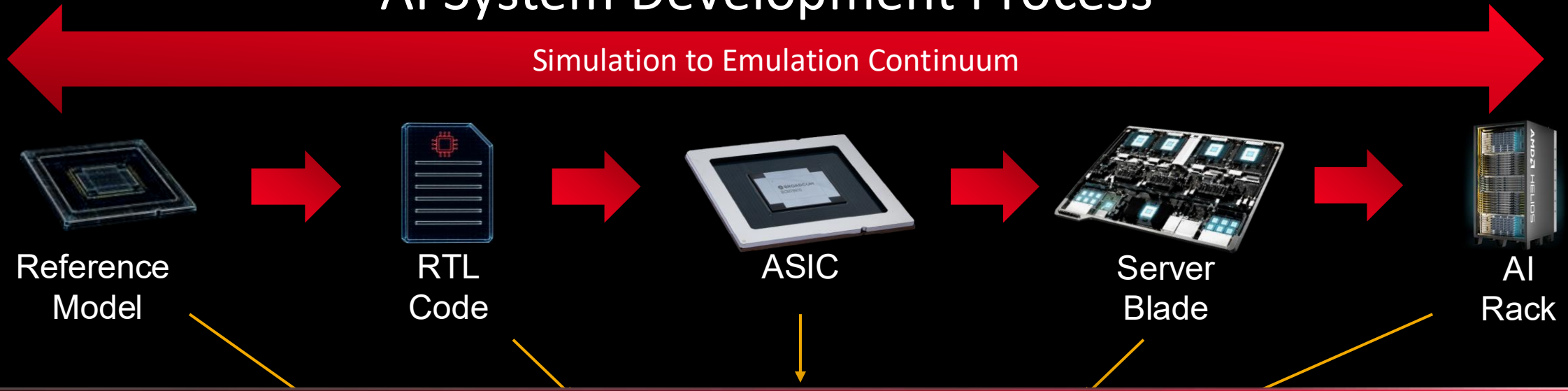


Workload Centricity

Efficiency

Tenets in Action

AI System Development Process

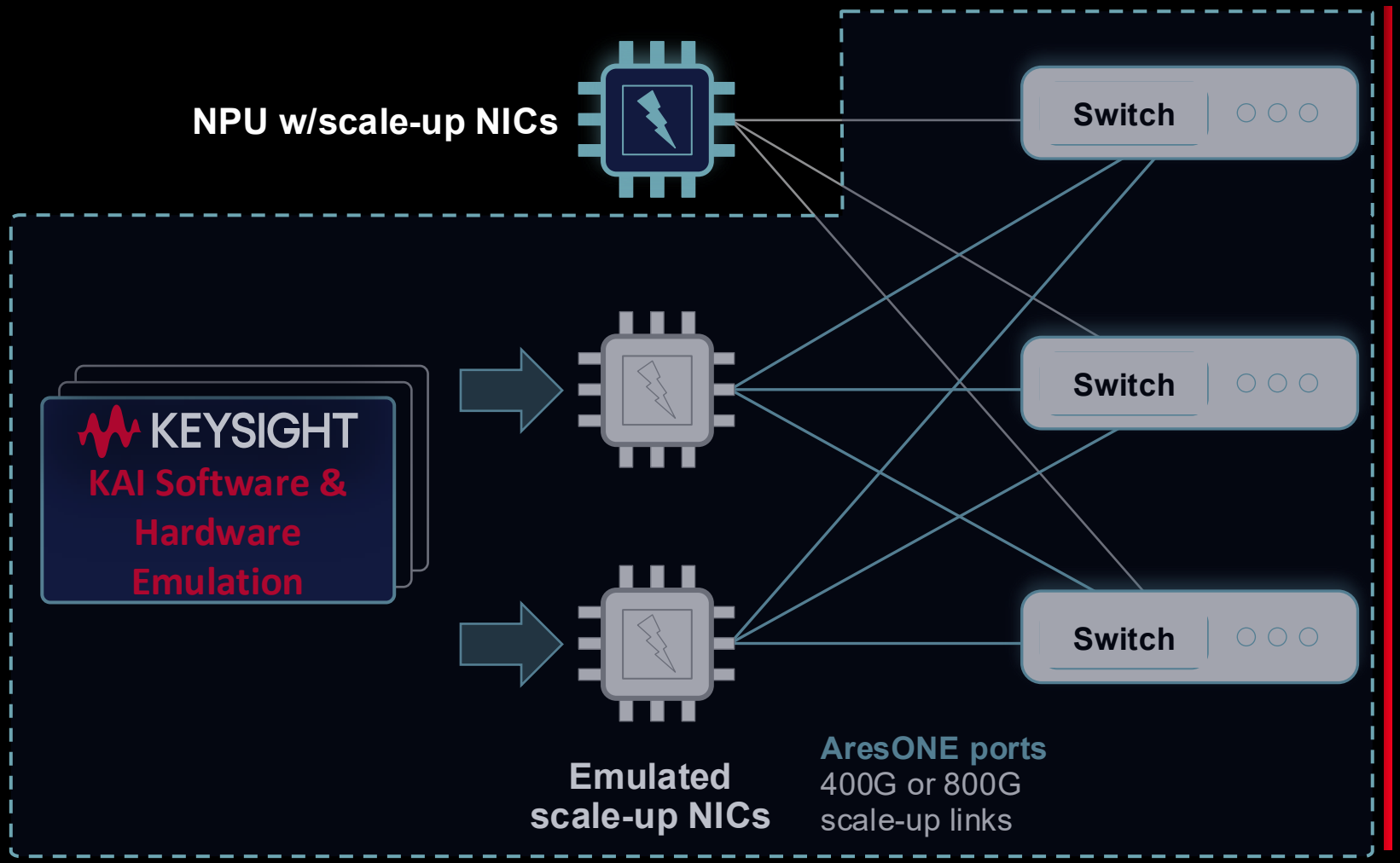


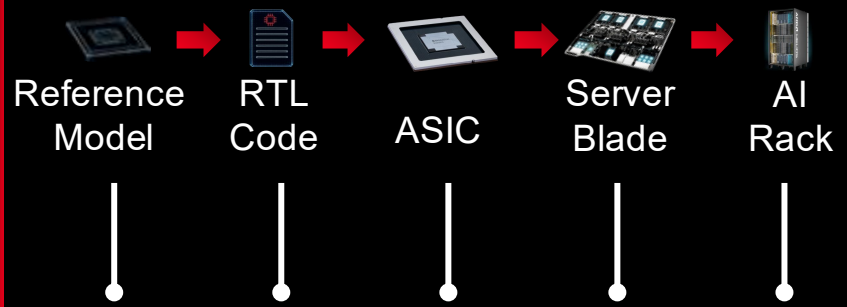
Reuse Upstream Work, Unified Source of Truth, Standardized Benchmarks, System Design Mindset At Every Development Step
— Emulate the rest of the AI System at every phase
from the first line of reference code to deploying AI Racks

* Integrations with Synopsis, Cadence, Siemens

Image credits: NVIDIA, AMD, Broadcom

Tenets in Action – System Thinking

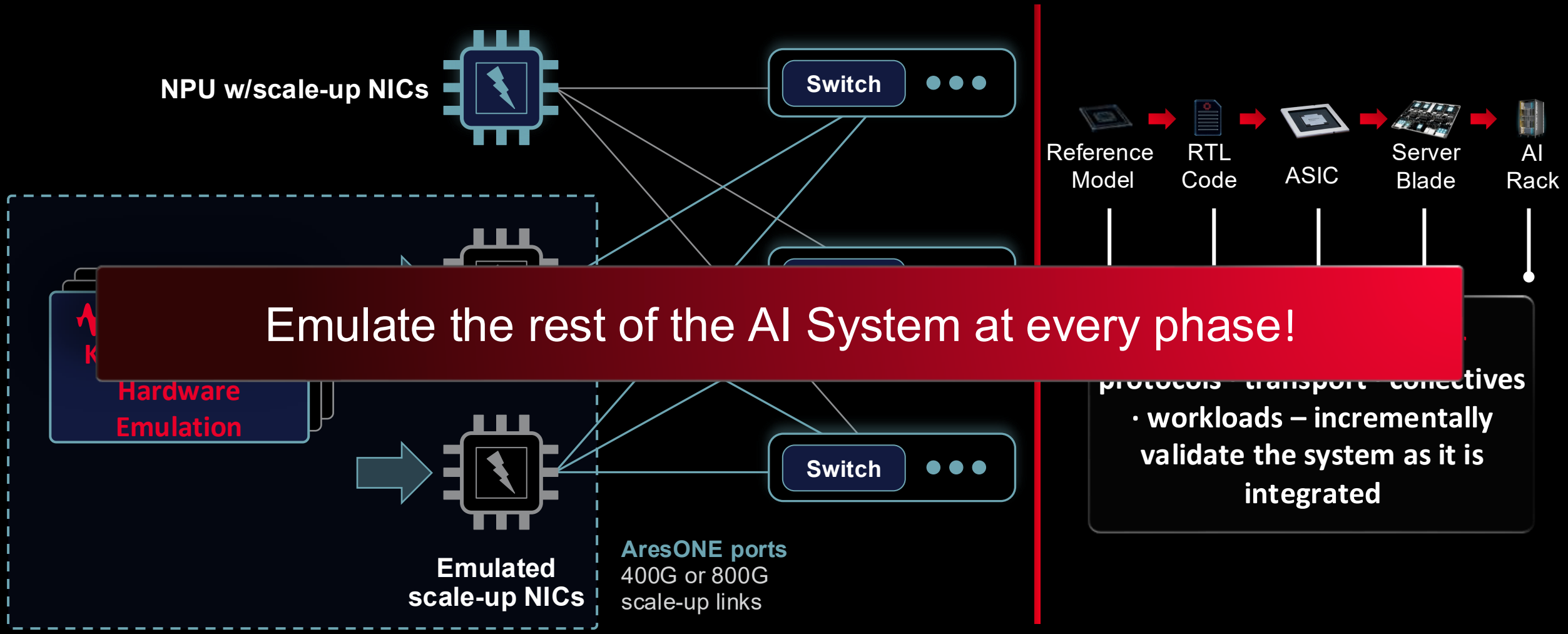




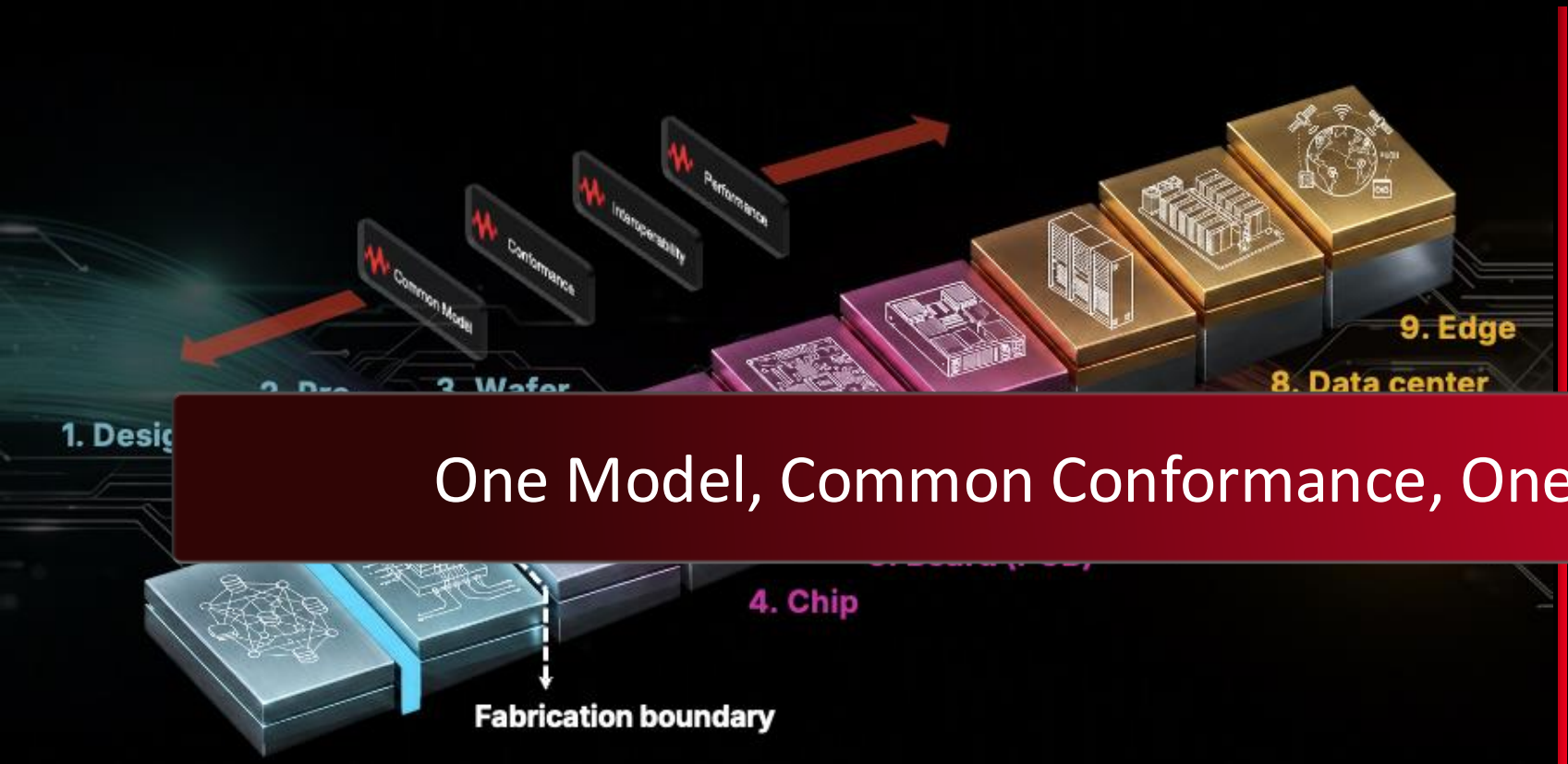
KAI Data Center Builder

protocols · transport · collectives · workloads – validate systems and scale early in the design

Tenets in Action – System Thinking



Tenets in Action – Continuum

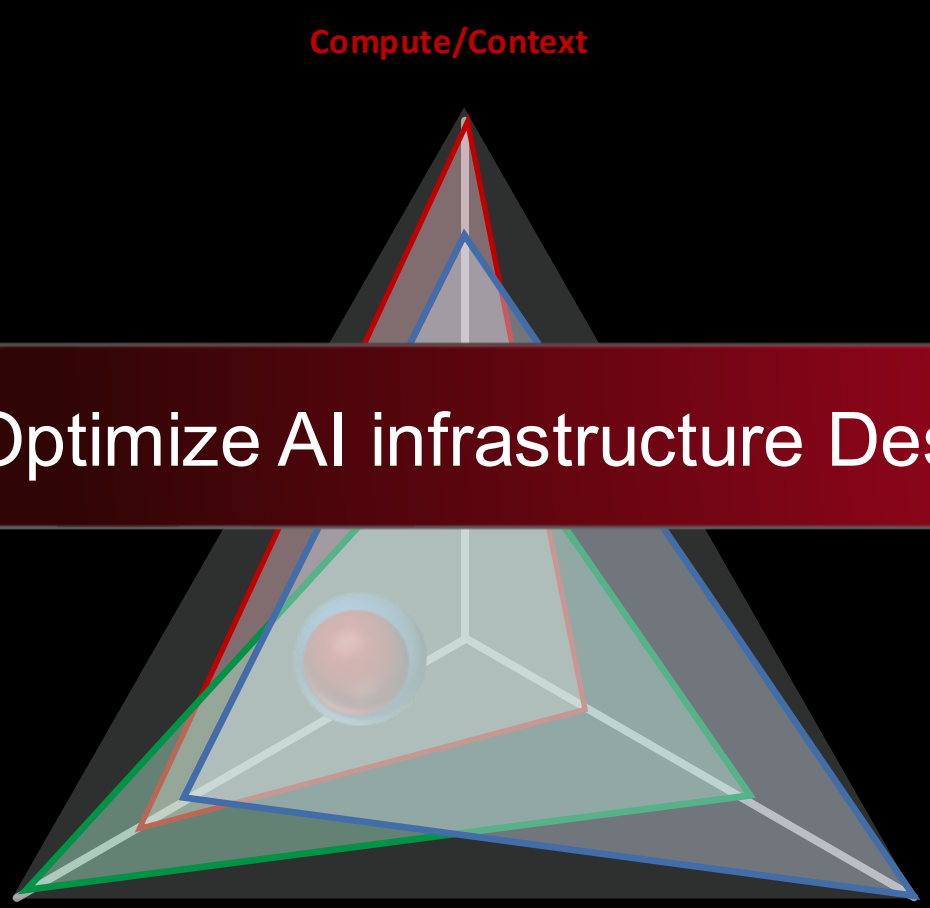


Reference Model → RTL Code → ASIC → Server Blade → AI Rack

One Model, Common Conformance, One Benchmark!

· one pass/fail bar, applied identically at every development stage

Tenets in Action – Workload Centricity



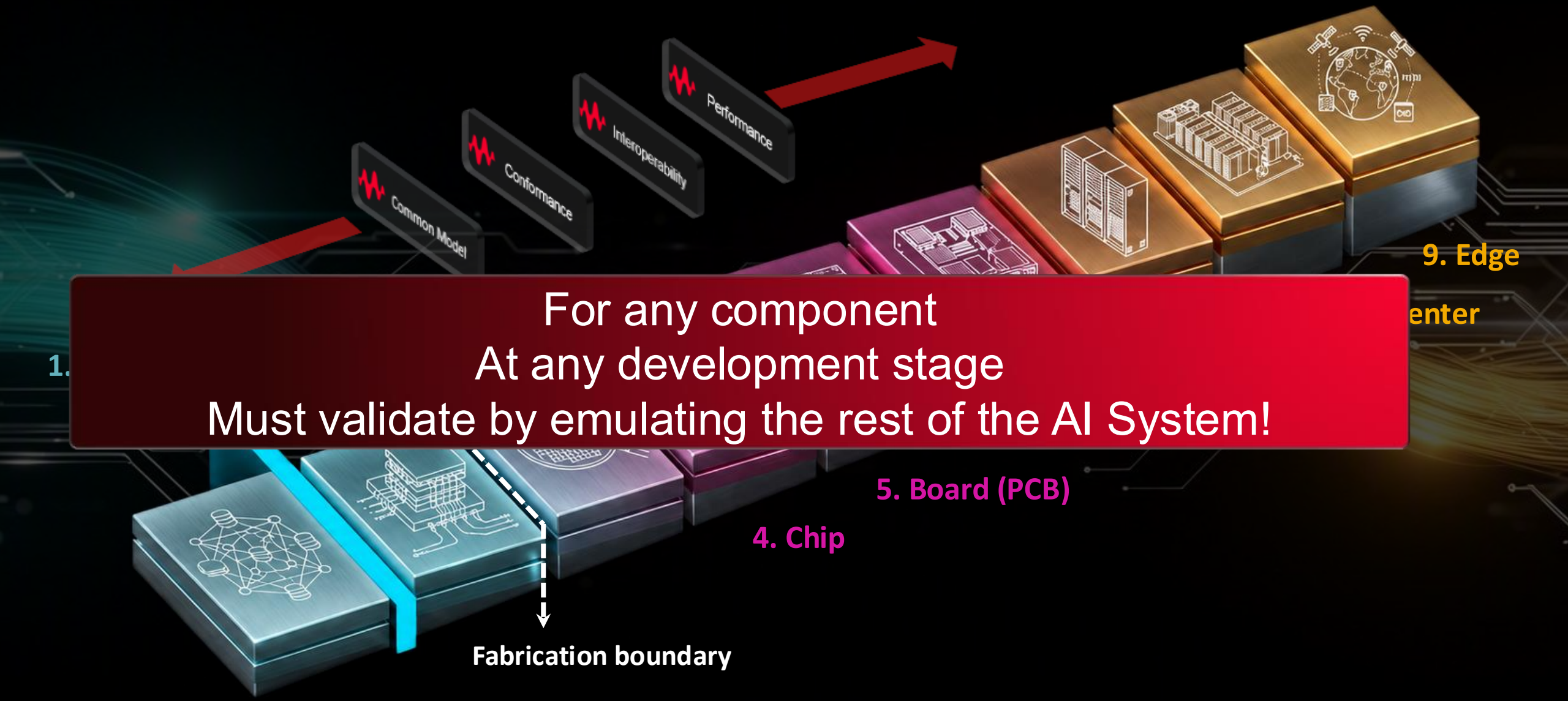
Reference Model → RTL Code → ASIC → Server Blade → AI Rack

Optimize AI infrastructure Design for Your Workloads!

KAI Inference Builder

emulate any AI workload · hardware/software co-design · optimize – emulate your workloads at every stage

Scaling AI: Design to Deployment Continuum



KEYSIGHT

Emulate Anything
Optimize Everything!